

Emily Bedolla Montes

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EDUCATION

Harmony School of Innovation, Fort Worth, TX

Expected Graduation: May 2026

High School Diploma with Distinction, *Computer Science Pathway*

GPA: 3.96 UW / 4.60 W

Relevant Coursework: *AP Computer Science Principles, AP Computer Science A, AP Physics C, AP Calculus AB, Dual Credit OnRamps Statistics*

SKILLS

- **Technical Skills:** Proficient in Java, Python, and Javascript; Familiar with CAD (Fusion 360, Onshape); Exposed to/learning SQL and R
- **Soft Skills:** Effective communication (oral and written), time management, public speaking, leadership, team collaboration, critical thinking, problem-solving
- **Credentials:** PCEP – Certified Entry-Level Python Programmer, Java IT Specialist, Autodesk Certified User Certification – Fusion 360
- **Languages:** Fluent in English and Spanish (Write, Read, and Speak)

PROFESSIONAL & TECHNICAL EXPERIENCE

Child Poverty Action Lab (CPAL), Dallas, TX

June 2025 - August 2025

Full Stack Development Intern, Full-Time

- Engineered enhancements to the Birth Rate Explorer platform, improving UI/UX responsiveness and backend data retrieval efficiency.
- Diagnosed and resolved backend bugs using JavaScript and React, increasing data accuracy and reducing load times.
- Designed and deployed a ZIP code search tool, enabling users to filter birth data by region and improving accessibility.
- Collaborated cross-functionally with developers and interns; presented work to executive leadership including CEO and CTO.

Code2College, Austin, TX (Remote)

August 2024 - Present

Participant - Software Development Essentials I

- Completed a 9-week immersive course covering Python programming, object-oriented design principles, and introductory web development.
- Built multiple Python-based projects applying intermediate concepts such as loops, functions, classes, and file handling.
- Participated in professional development workshops focused on resume writing, time management strategies, and building a technical portfolio.

Participant - Elite 101

- Selected for a competitive internship readiness program emphasizing Git version control, Jira ticketing systems, Miro-based team collaboration, and software testing methodologies.
- Designed and implemented a Python-based Chat Bot capable of simulating interactive conversations with users.
- Led a team in developing a virtual game prototype during DRW's STEM Case Competition; awarded 1st place among peer teams for innovation and execution.
- Earned a paid internship at CPAL by successfully completing a technical interview and showcasing project work.

Participant - Full Stack Development II

- Selected for a competitive upper-level software engineering program focused on advanced full stack development.

- Building a fully functional e-commerce web application using React for the front end, Node.js for the back end, and PostgreSQL for database management.
- Implementing user authentication, dynamic product listings, and secure checkout functionality.
- Strengthening skills in RESTful API design, responsive UI development, and SQL query optimization.

PROJECTS

E-Commerce Website

September 2025 – Present

- Building a full-stack e-commerce site using React, Node.js, Express, and PostgreSQL.
- Created dynamic product listings, user authentication, and shopping cart functionality.
- Designed RESTful APIs to manage users, products, and orders.
- Structured the database to support inventory tracking and order history.
- Testing and refining features to improve performance and user experience.

Birth Rate Explorer

July 2025 – August 2025

- Developed and enhanced a data-driven web application for visualizing birth rate statistics using React and Mapbox GL JS.
- Researched and debugged Mapbox API integration issues, resolving a rendering bug that affected geospatial data overlays and improving map responsiveness across devices.
- Refactored CSS architecture to adopt modular styling conventions, streamline layout consistency, and improve cross-browser UI performance.
- Updated state management logic and variable structures to accommodate new birth rate datasets; conducted unit testing and data validation to ensure seamless integration.
- Designed and implemented a reusable React component that enables ZIP code-based map filtering, allowing users to dynamically query and visualize localized birth data.

Python-Based Chat Bot

January 2025 – March 2025

- Designed and developed an interactive chatbot using Python that simulates human-like conversation through structured dialogue flow and conditional logic.
- Implemented core features including keyword recognition, branching responses, and input validation to guide users through dynamic conversation paths.
- Incorporated string manipulation techniques and control structures to personalize responses based on user input.
- Conducted iterative testing to refine dialogue accuracy, eliminate edge-case errors, and ensure consistent user experience.

Finances Tracker

April 2024

- Designed and built a personal finance tracking program using Python to help users monitor income and expenses.
- Implemented core features including input validation, categorized expense logging, and dynamic balance updates.
- Used lists, functions, and conditionals to organize data and support user interaction through a text-based interface.
- Tested program functionality with sample data and refined logic to meet Create Task requirements.
- Documented development process, including purpose, functionality, and abstraction, in alignment with AP CSP guidelines.

Self-Watering Plant System

November 2023 – January 2024

- Engineered an automated irrigation system using Arduino, moisture sensors, and a water pump.
- Programmed logic to detect soil dryness and activate watering cycles, conserving water and optimizing plant health.
- Designed a mobile interface to monitor moisture levels and manually trigger watering remotely.

Automatic Chicken Coop Door

June 2023

- Built a motorized coop door system using Arduino, servo motors, and light sensors to automate opening/closing based on daylight.
- Programmed safety overrides and manual controls to ensure animal welfare and system reliability.
- Designed a weather-resistant enclosure and tested hardware durability under outdoor conditions.

LEADERSHIP & COMMUNITY INVOLVEMENT

GAINS (Girls Advancing in Stem), Fort Worth, TX

August 2024 – Present

Founding officer, Head of Finances and Events

- Co-founded HSI-FW's first GAINS chapter to mentor 20+ girls in STEM and support passion projects.
- Organizing fundraisers and events to support club initiatives and coordinated logistics for Carnegie Mellon University conference trip.
- Sharing curated resources including scholarships, internships, and research opportunities.
- **Spent 1 and a half hours per week for about 30 weeks on this activity.**

UIL Computer Science, Fort Worth, TX

August 2024 – Present

Member

- Competing in coding competitions focused on data structures, algorithms, and problem-solving under time constraints, including a competition called Code Quest which is hosted by Lockheed Martin.
- Completed Advent of Code series to strengthen problem-solving in Java.
- Developed strong programming skills in Java and honed logical thinking through individual and team-based challenges.
- Demonstrated proficiency in competitive programming concepts, including recursion, dynamic programming, and graph theory.
- **Spent 2 and a half hours per week for about 30 weeks on this activity.**

HSI Student Council, Fort Worth, TX

August 2023 – Present

Vice President

- Directing 50+ members in meetings and organized school events, activities, and fundraisers such as Homecoming, Candy Jar, Hallapalooza, Talent Show, and Spring Formal.
- Managed council finances through budgeting spreadsheets and fundraising strategies, raising over \$3K for student activities.
- Promoting school spirit and engagement, contributing 60+ hours of community service.
- **Spent 3 hours per week for about 30 weeks on this activity.**

Aerial Drone Competition, Fort Worth, TX

August 2023 – Present

Lead Software Engineer, Team Manager

- Leading a team of software engineers to come up with a game-plan/strategy and design and code the autonomous mission for 15+ regional and national competitions.
- Programming drone behavior using Python, integrating sensors (range, color), gyroscope data, and flight controls (roll/pitch/yaw/throttle).
- Managing team logistics, including competition schedules, task assignments, and technical documentation for 75+ practices.
- Mentoring junior members in drone programming and competition strategy.
- Earned 7 awards in autonomous programming and other categories and qualified for Nationals two consecutive years.
- Achieved a top global autonomous rank; recognized by leaders in the REC (Robotics Education & Competition Foundation) for exemplar logbook which is now being used as an example logbook for new Aerial Drone teams.
- **Spent 4 hours per week for about 30 weeks on this activity.**

Hispanic Honor Society, Fort Worth, TX

August 2023 – Present

Member

- Supporting cultural events including Hispanic Heritage Festival and Dia de Los Muertos altar.
- Shared heritage through educational outreach and event planning.
- Contributed 10+ hours of community service.
- **Spent 1 hour per week for about 30 weeks on this activity.**

HSI Varsity Girls Volleyball Team, Fort Worth, TX

August 2022 – November 2024

Co-Captain

- Led team through 90+ practices and 24 competitive matches, fostering discipline and sportsmanship.
- Developed strategic plays and coordinated in-game adjustments with coaching staff.
- Guided team to playoff qualification through consistent performance and leadership.
- **Spent 6 hours per week for about 10 weeks on this activity.**

AWARDS/HONORS

- AP Scholar with Distinction
- 98th Percentile SAT Scorer in the state of Texas
- School Science Fair 1st Place Winner
- Aerial Drone Competition Nationals Communications Mission Champion
- Aerial Drone Competition Career Readiness Award
- Aerial Drone Competition Coding Award
- Aerial Drone Competition Airmanship Award
- Aerial Drone Competition Judges Award
- Aerial Drone Competition Excellence Award
- Aerial Drone Competition Flight Operations Award
- 100+ Volunteer Hours
- National Junior Honor Society
- A Honor Roll
- National First-Generation Recognition Program
- National Hispanic Recognition Program
- National School Recognition Program
- QuestBridge College Prep Scholar
- QuestBridge Match Finalist